JUnit Exercises 1–4

Setup, basic tests, assertions, AAA pattern with JUnit 4

Exercise 1: Setting Up JUnit

pom.xml (JUnit dependency)

pom.xml

```
<dependencies>
    <dependency>
        <groupId>junit</groupId>
        <artifactId>junit</artifactId>
        <version>4.13.2</version>
        <scope>test</scope>
    </dependency>
</dependencies>
```

Starter test class

src/test/java/CalculatorTest.java

```
import org.junit.Test;

import static org.junit.Assert.assertEquals;

public class CalculatorTest {

    @Test
    public void sampleTest() {
        assertEquals(4, 2 + 2);
    }
}
```

Exercise 2: Writing Basic JUnit Tests

Class under test

src/main/java/Calculator.java

```
public class Calculator {

    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }
}
```

Basic tests

src/test/java/CalculatorTest.java

```
import org.junit.Test;

import static org.junit.Assert.assertEquals;

public class CalculatorTest {

    private final Calculator calculator = new Calculator();

    @Test
    public void testAdd() {
        assertEquals(5, calculator.add(2, 3));
    }

    @Test
    public void testSubtract() {
        assertEquals(1, calculator.subtract(3, 2));
    }
}
```

Exercise 3: Assertions in JUnit

src/test/java/AssertionsTest.java

```
import org.junit.Test;

import static org.junit.Assert.*;

public class AssertionsTest {

    @Test
    public void testAssertions() {
        // Assert equals
        assertEquals(5, 2 + 3);

        // Assert true
        assertTrue(5 > 3);

        // Assert false
        assertFalse(5 < 3);

        // Assert null
        assertNull(null);

        // Assert not null
        assertNotNull(new Object());
    }
}
```

Exercise 4: Arrange-Act-Assert Pattern, Setup and Teardown

src/test/java/CalculatorAAATest.java

```java
import org.junit.After;
import org.junit.Before;
import org.junit.Test;

import static org.junit.Assert.assertEquals;

public class CalculatorAAATest {

    private Calculator calculator;

    @Before
    public void setUp() {
        // Arrange (shared fixture)
        calculator = new Calculator();
    }

    @Test
    public void testAdd() {
        // Arrange
        int a = 2;
        int b = 3;

        // Act
        int result = calculator.add(a, b);

        // Assert
        assertEquals(5, result);
    }

    @Test
    public void testSubtract() {
        // Arrange
        int a = 10;
        int b = 4;

        // Act
        int result = calculator.subtract(a, b);

        // Assert
        assertEquals(6, result);
    }

    @After
    public void tearDown() {
        calculator = null;
    }
}
```